Hardware accelerated telemetry data compression platform

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12th European CubeSat Symposium, 26/11/2021

Current State

- High throughput and deep space CubeSats
- Ground infrastructure costs
- Limited ground and on-board resources

Approach

Data Compression

POCKET+ Standard

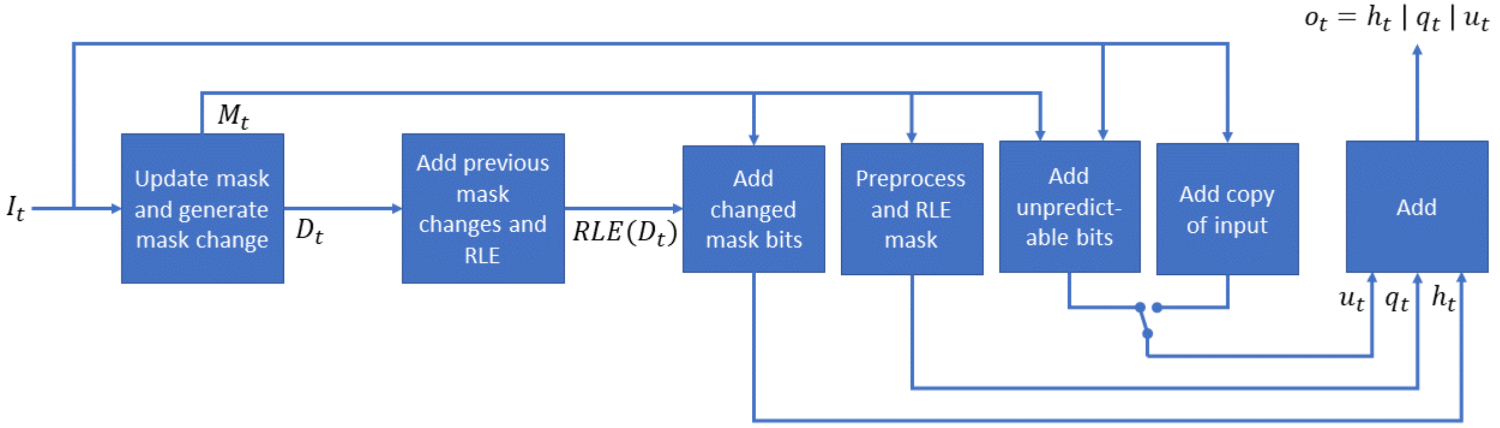
Hardware Acceleration

POCKET+

- Based on ESA patented compression algorithm “POCKET”
- CCSDS 124.0-W-4 draft standard (November 2020)
  “Robust Compression of Fixed-Length Housekeeping Data”
- Differential and lossless compression algorithm
- Free and Open-Source (AGPL-3.0) C++ implementation
  https://github.com/visionspacetec/PocketPlus

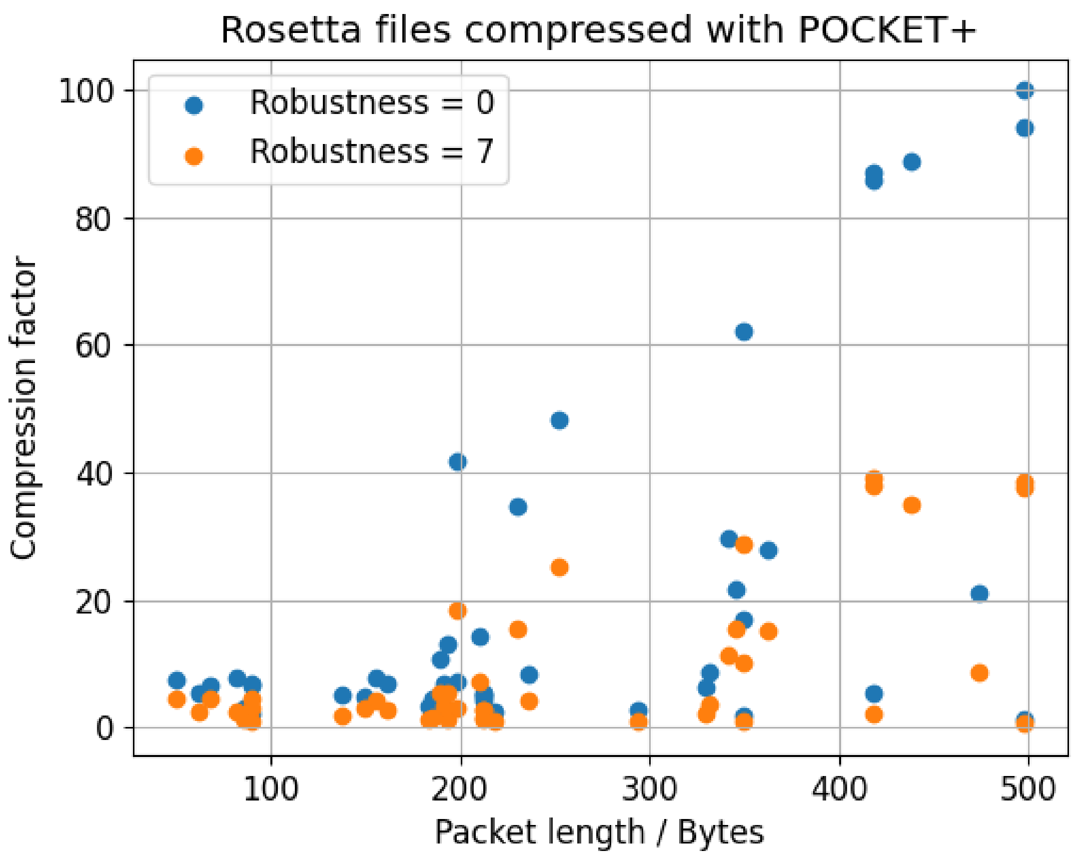
POCKET+ Algorithm

- Mask generation
- Robustness level
- Extract changes
- Run length encoding
- Uncompressed flag
- New mask flag



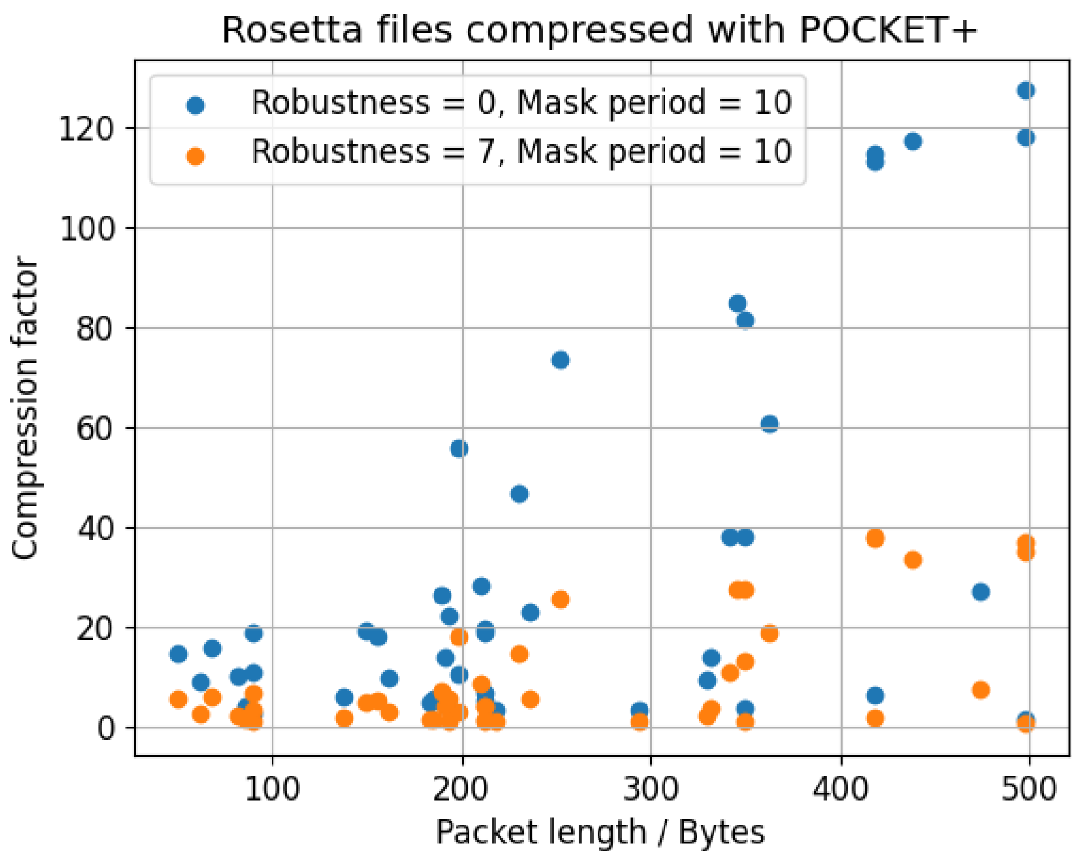
POCKET+ Compression Performance

- Rosetta, VEX housekeeping data
- Robustness affects performance
- Best for large packets
- Factor 14 for no robustness
- Factor 6.5 for max. robustness



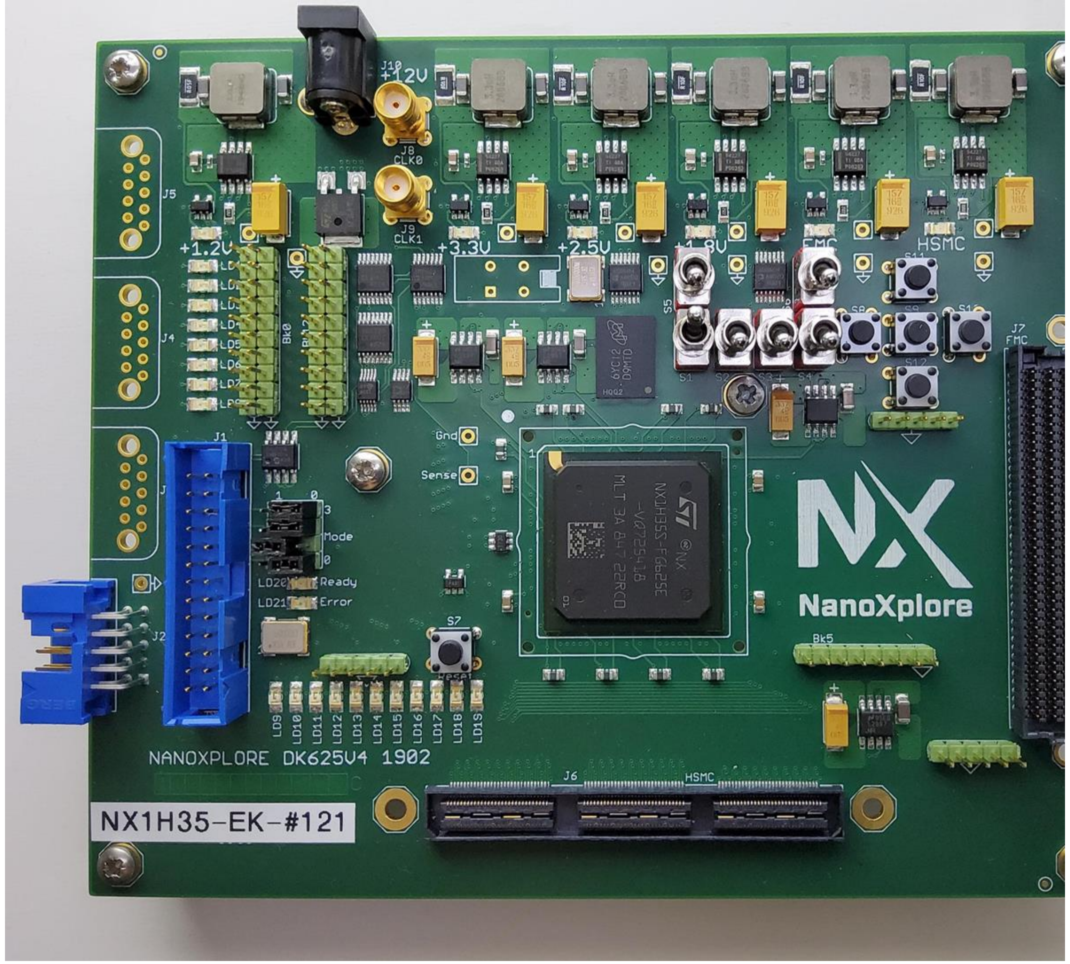
POCKET+ Compression Performance

- Using algorithm feature to improve performance
- Reset mask periodically
- Increased compression factor from 14 to 23.8 @ R=0
- Algorithm can be adaptive



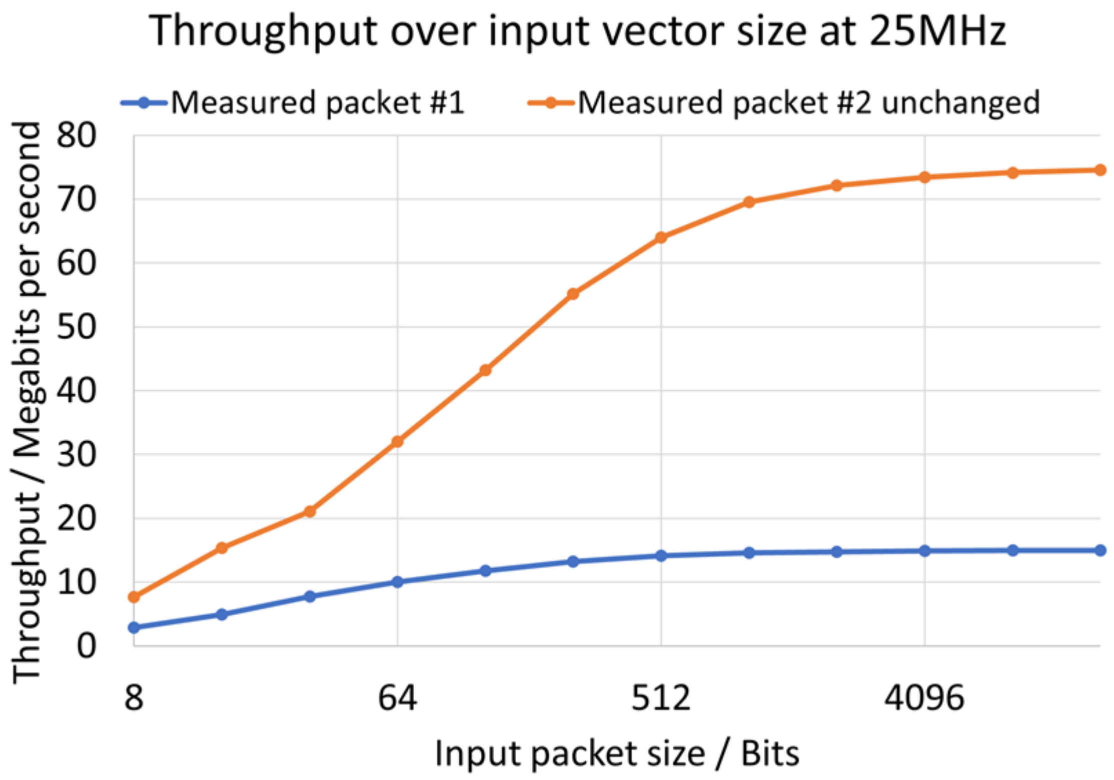
NanoXplore NG-Medium

- Radiation hardened FPGA
- European product (ITAR-free)
- 34272 4-LUTs
- 56 ECC protected BRAMs
- DSP blocks
- Development kits available



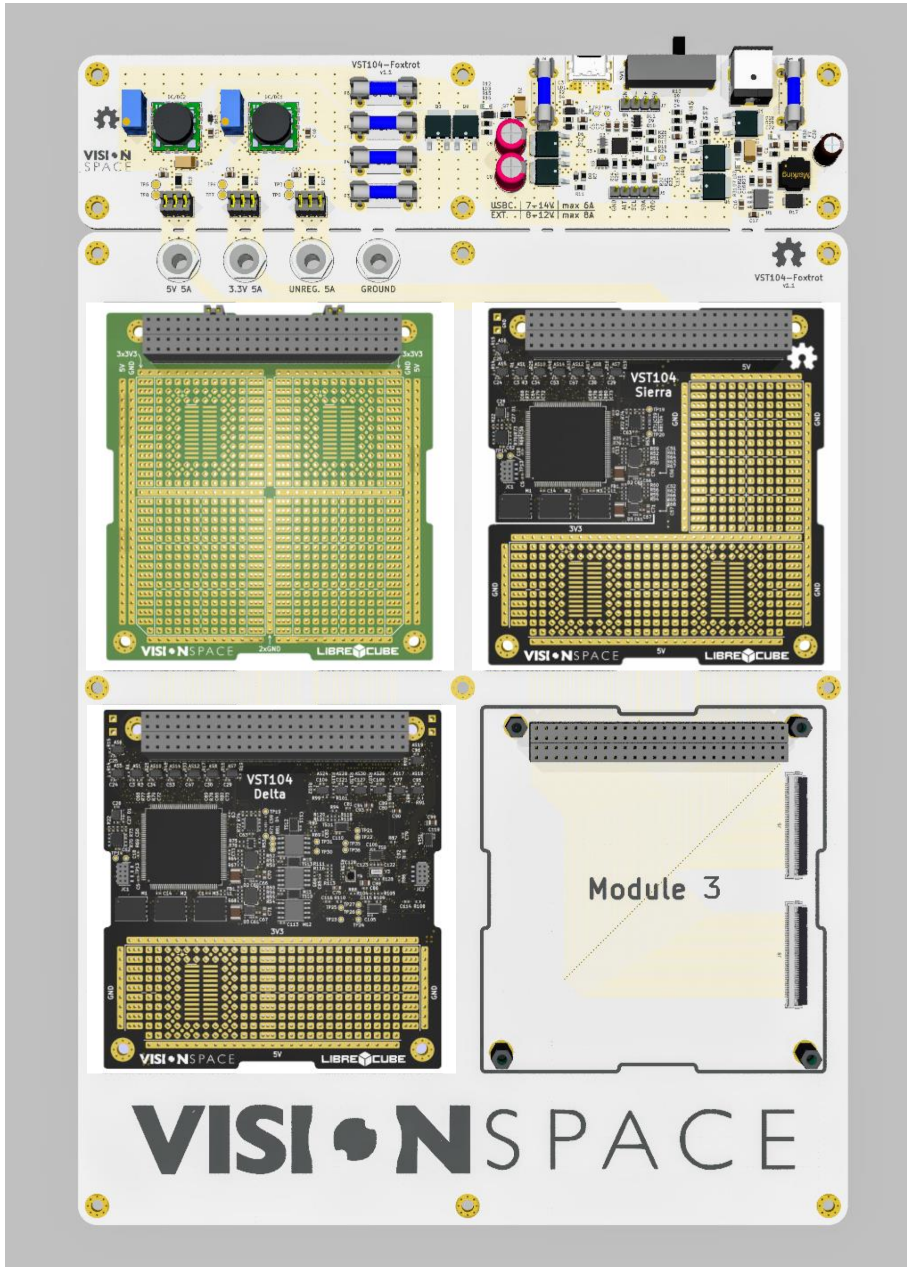
POCKET+ on NG-Medium

- First-time FPGA implementation
- Evaluated IP core performance
- 3-8 Mbit/s @ 8 bits
- 15-75 Mbit/s @ 16384 bits
- Used 30% of LUTs and BRAMs



VST104

- CubeSat on-board computer series
- Prototyping – “Zero”
- Single MCU – “Sierra”
- Dual MCU – “Delta”
- Flatsat – “Foxtrot”
- Free and Open Hardware (CERN-OHL-S) https://github.com/visionspacetec/VST104



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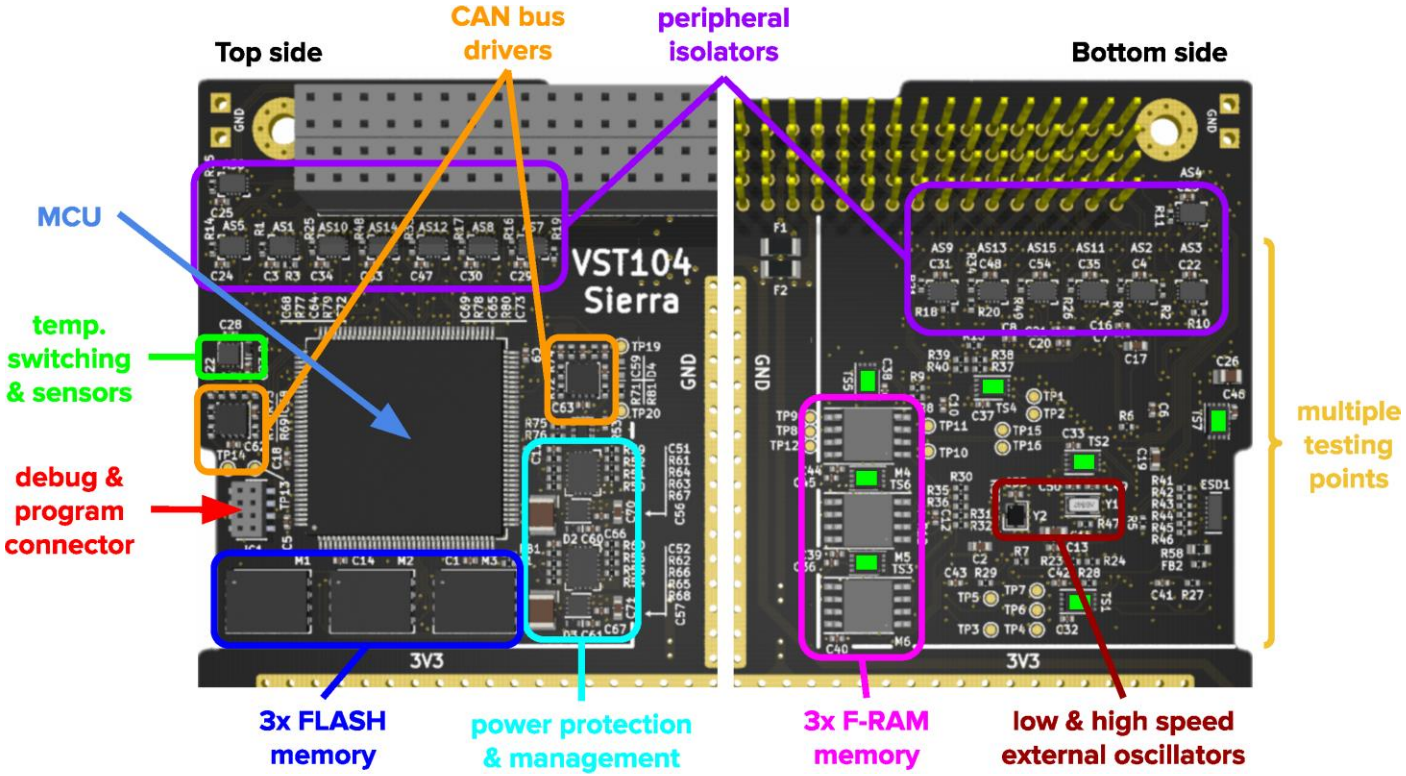
VST104 – Board Sierra

- ARM Cortex M4
- Trippel redundant memory
- 256 Mbit Flash
- 2 Mbit F-RAM
- Power protection
- Redundant CAN bus
- Peripheral isolators



Prust

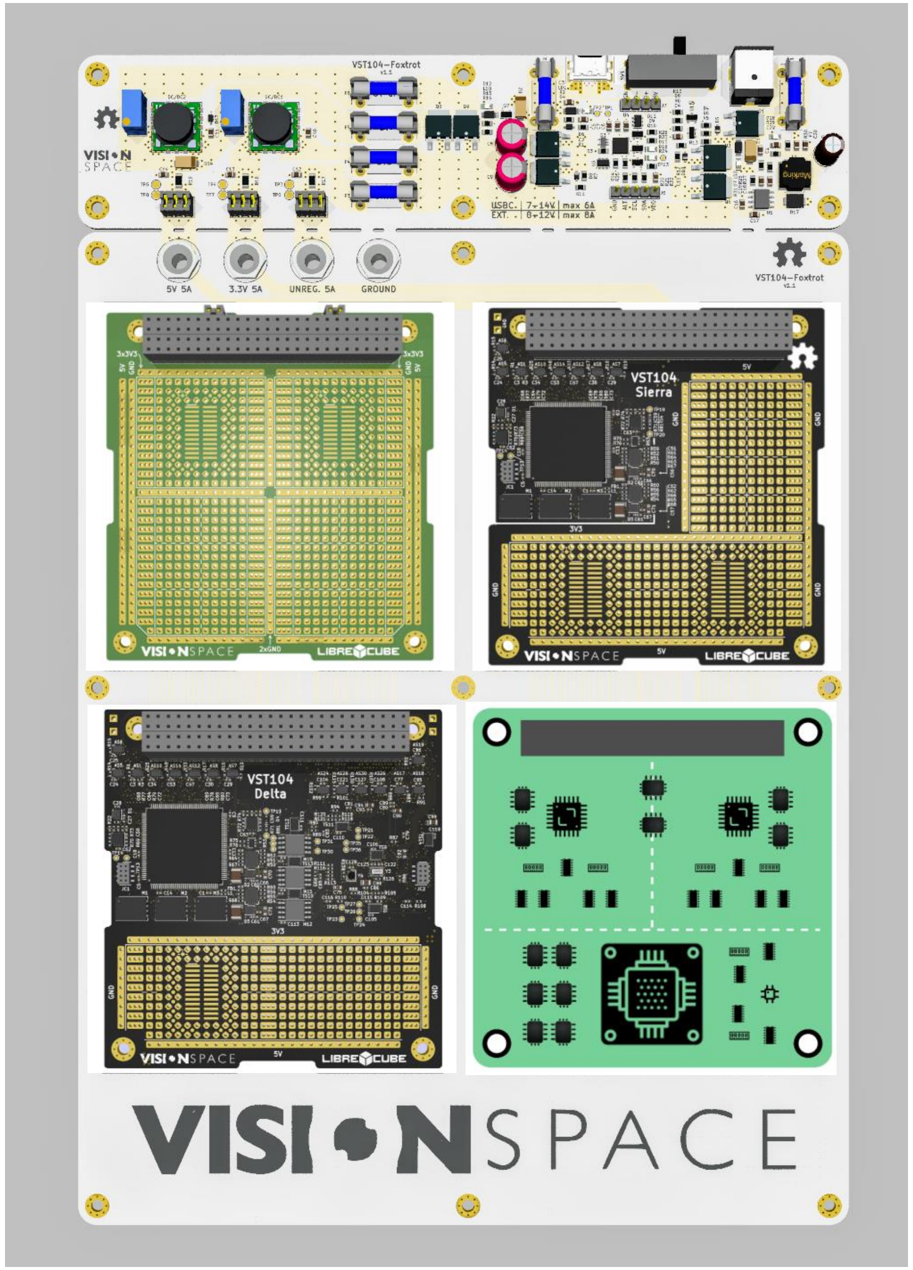
- On-board software stack
- ECSS PUS-C standard
  "Telemetry and Telecommand Packet Utilization"
- Rust programming language
- FreeRTOS integration
- Free and Open-Source (AGPL-3.0)
  https://github.com/visionspacetec/Prust

Putting it all Together

- PC104 prototyping board
- VST104 On-board computer series
- Flatsat board
- TM/TC firmware
- VisionSpace SLE provider

https://github.com/visionspacetec/sle-provider

- C2LOCO - EGS-CC tailored for CubeSats



Outlook

- Update POCKET+ based on to-be-released standard
- Add a high throughput FPGA interface
- VST104 board with integrated NG-Medium FPGA
- More Rust libraries, including a timeline-based planner and executor

Thank you!

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